

INSTRUCTIONS TO THE CANDIDATES

- ❖ Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
- ❖ Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
- ❖ Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
- ❖ Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
- ❖ The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.

Non-adherence with this instruction shall result in candidate being booked under malpractice.

- ❖ Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
- ❖ Candidate should write only question number in the margin.
- ❖ Candidate should return the answer booklet to the invigilator before leaving the examination hall.
- ❖ **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- ❖ Invigilators should ensure that the above instructions are strictly followed by the candidates.
- ❖ After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- ❖ Do not crumple or cut the sides of the sheets.
- ❖ Do not write, mark or damage the timing tracks or Barcode.

3 KKT

$$\text{Max } Z = 10x_1 + 4x_2 - 2x_1^2 - x_2^2$$

Subject to constraints.

$$- 2x_1 + x_2 \leq 5 \Rightarrow 2x_1 + x_2 - 5 \leq 0$$
$$x_1, x_2 \geq 0$$

1 constraint.

$$h = 2x_1 + x_2 - 5 \quad \lambda \geq 0$$

Lagrangian function.

$$L(x_1, x_2, \lambda) =$$

$$10x_1 + 4x_2 - 2x_1^2 - x_2^2 - \lambda(2x_1 + x_2 - 5)$$

Find

$$\frac{\partial L}{\partial x_1} = 0$$

$$\frac{\partial L}{\partial x_2} = 0$$

$$L = 10x_1 + 4x_2 - 2x_1^2 - x_2^2 - \lambda(2x_1 + x_2 - 5)$$

$$\frac{\partial L}{\partial x_1} = 10 - 4x_1 - 2\lambda = 0 \quad - (1)$$

$$\frac{\partial L}{\partial x_2} = 4 - 2x_2 - \lambda = 0 \quad - (2)$$

$$\lambda(h) = 0 \Rightarrow \lambda(2x_1 + x_2 - 5) = 0 \quad - (3)$$

Since there is 1 λ constraint variable there are 2 scenarios.

$$\cdot \lambda = 0$$

$$\cdot \lambda \neq 0$$

$$\rightarrow \lambda = 0$$

if $\lambda = 0$ then when substituted in (1) and (2)

$$10 - 4x_1 = 0$$

$$4 - 2x_2 = 0$$

$$\Rightarrow 10 = 4x_1$$

$$x_1 = \frac{10}{4} = \frac{5}{2}$$

$$\Rightarrow 4 = 2x_2$$

$$x_2 = 2$$

$$x_1 = \frac{5}{2} \quad x_2 = 2$$

Check if it satisfies given constraints.

$$2(x_1) + (x_2) - 5 \leq 0$$

$$2\left(\frac{5}{2}\right) + 2 - 5 \leq 0$$

$$2 \not\leq 0$$

Not Satisfies

reject

$$\rightarrow \lambda \neq 0$$

if $\lambda \neq 0$ then when substituted in ③

$$\lambda(2x_1 + x_2 - 5) = 0$$

as $\lambda \neq 0$ then $2x_1 + x_2 - 5 = 0$

rearrange ① ② and obtained ③

$$-4x_1 + 0x_2 - 2\lambda + 10 = 0$$

$$0x_1 - 2x_2 - \lambda + 4 = 0$$

$$2x_1 + x_2 - 0\lambda - 5 = 0$$

$$\begin{array}{r} -4x_1 + 0x_2 - 2\lambda + 10 = 0 \\ + \quad 4x_1 + 2x_2 - 0\lambda - 10 = 0 \\ \hline 2x_2 - 2\lambda = 0 \end{array}$$

$$\boxed{x_2 = \lambda}$$

$$-2x_2 - \lambda + 4 = 0$$

$$-2\lambda - \lambda + 4 = 0$$

$$-3\lambda = -4$$

$$\boxed{\lambda = \frac{4}{3}} \Rightarrow \boxed{x_2 = \frac{4}{3}}$$

$$2x_1 + x_2 - 5 = 0$$

$$2x_1 + \frac{4}{3} = 5$$

$$6x_1 + 4 = 15$$

$$6x_1 = 11$$

$$\boxed{x_1 = \frac{11}{6}}$$

$$\boxed{x_1 = \frac{11}{6}, x_2 = \frac{4}{3}, \lambda = \frac{4}{3}}$$

check if it satisfies constraints

$$2(x_1) + x_2 - 5 \leq 0$$

$$2(x_1) + (x_2) - 5 \leq 0$$

$$2\left(\frac{11}{6}\right) + \left(\frac{4}{3}\right) - 5 \leq 0$$

$$\frac{11+4}{3} - 5 \leq 0$$

$$\frac{15-15}{3} \leq 0$$

$$0 \leq 0$$

Satisfies.

Find Z

$$10x_1 + 4x_2 - 2x_1^2 - x_2^2$$

$$10\left(\frac{11}{6}\right) + 4\left(\frac{4}{3}\right) - 2\left(\frac{11}{6}\right)^2 - \left(\frac{4}{3}\right)^2$$

$$\frac{110}{6} + \frac{16}{3} - 2\left(\frac{121}{36}\right) - \left(\frac{16}{9}\right)$$

$$\frac{6(110) + 12(16) - 2(121) - 4(16)}{36}$$

$$\frac{660 + 192 - 242 - 64}{36} = \frac{546}{36} = \frac{91}{6}$$

$$Z = \frac{91}{6}$$

6

Genetic algorithm.

$$\text{function } f(x) = x^2$$

Interval [0, 31]

13, 9, 5, 16

String No	Interval Points	Binary	f(x)	f(x)/Avg.	Accum cost
1	13	01101	169	1.3	1
2	9	01001	81	0.7	1
3	5	00101	25	0.2	0
4	16	10000	256	1.9	2

$$\text{Sum} = 531$$

$$\text{Max} = 256$$

$$\text{Avg} = 132.7$$

(CROSS OVER)

S.No	Initial Val	Cross over	Off Spring	Number	f(x)
1	01101	4	0100	12	144
4	10000 01001	4	10001	17	289
2	10000	2	01000	8	64
4	10000	2	00001	17	289

Sum

$$\text{Sum} = 786$$

$$\text{max} = 289$$

$$\text{Avg} = 196.5$$

Mutation

S.No	Offspring	Mutating point	Offspring after mutation	Number	f(x)
1	01100	00010	01110	12 14	196
4	10001	00010	00011	19	361
2	01000	00010	01010	10	100
4	00001	00010	10011	19	361

$$\text{Sum} = 1018$$

$$\text{max} = 361$$

$$\text{Avg} = 254.5$$

A The Concept of Non Linear programming is that. Not all the functions and constraints are given in a Linear form. There can be quadratic programming

B

Examples of quadratic programming

KKT algorithm

Beale's method

Wolfe's method

C

Quadratic programming and Linear programming problems are similar in a sort way. But the way we approach them is how we tackle the problem.

In Linear programming the Objective function and constraints are in Linear form.

In Quadratic programming the Objective function is in Quadratic form, but the constraint may or may not.

E

Cross over techniques.

The cross over technique is used in the genetic algorithm to see the mutations cross over and expand to see how far the expression can be made.

$$\begin{array}{|c|c|c|c|c|} \hline 1 & 0 & 0 & 1 & 1 \\ \hline 1 & 1 & 0 & 0 & 1 \\ \hline \end{array} \begin{array}{l} 2 \\ 2 \end{array} \Rightarrow \begin{array}{|c|c|c|c|c|} \hline 1 & 0 & 0 & 0 & 1 \\ \hline 1 & 1 & 0 & 1 & 1 \\ \hline \end{array}$$

2.

A

KKT optimisation.

In KKT optimization the objective function is given in a quadratic form and the constraints are given in linear form. The significance is that it optimizes the function with Lagrange's and linear constraints. The objective function is converted into a Lagrangian expression with its constraints. The Lagrangian expression is checked under conditions and the one suited is the answer.

D

The merits ACO

Merits:

- Finds the optimal path for Ants which leave its pheromones on its trail.
- Most used trail found
- Helps in navigating.

C

Genetic

Selection and crossover.

It is seen in Genetic algorithm.

- Optimal selection
- selection best value.
- mutation for the better
- Crossover into Linking with intervals

B

Beale's

- Optimal fraction
- Constraints
- Basic non Basic variable
- Finding the optimal non Basic variable
- How much can x_2 be increased
- Entry variable
- Multiple iterations
- Find all values of Basic and non Basic variables
- Find optimal fraction
- Extra time in Exams



